

Analysis of ECG Signals Using Laplace Transform for Portable Heart Pulse Monitoring

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Abstract - Electrocardiogram (ECG) signals are essential for monitoring the electrical activity of the heart and provide valuable information about cardiac function (eMedicineHealth, 2025; Hal, 2010). This study analyzes ECG signals in both the time and frequency domains using the Laplace transform as a signal processing tool (Antony Raj, 2025). The ECG data are obtained from recorded signal sources. Time-domain analysis is used to examine the waveform shape and fundamental features of the ECG signal, while frequency-domain analysis identifies dominant frequency components associated with heart activity (MDPI, 2022). The Laplace transform is applied to represent the ECG signal in the transform domain and to describe its behavior mathematically. By combining time-, frequency-, and Laplace-domain analyses, this study offers a clearer understanding of ECG signal characteristics. The results demonstrate that basic signal processing techniques can be effectively applied to biomedical signals and can serve as a practical reference for students learning about signal spectra and signal processing.

Keywords – *Electrocardiogram (ECG), Laplace Transform, Arduino, Filtering, Normalization*

I. INTRODUCTION

The heart is a vital organ that generates electrical signals responsible for maintaining proper blood circulation in the human body. These electrical activities can be measured and recorded using an Electrocardiogram (ECG), a non-invasive diagnostic tool commonly used to monitor heart rate, rhythm, and overall cardiac condition (Hal, 2010; Antony Raj, 2025). An ECG records the electrical potential differences produced during the cardiac cycle and displays them as a waveform over time (MDPI, 2022).

An ECG signal is composed of several important components, namely the P wave, QRS complex, and T wave. Each part of the waveform represents a specific electrical event in the heart, such as atrial depolarization and ventricular repolarization (Hal, 2010). Studying these components in the time domain allows direct observation of signal amplitude and duration, which is useful for identifying irregularities in heart rhythm (MDPI, 2022). However, ECG signals are classified as non stationary signals, meaning their characteristics change over time. Because of this, time-domain analysis alone may not always provide complete information about the signal behavior. Frequency-domain analysis is therefore used to examine the spectral components of ECG signals and determine how signal energy is distributed across different frequencies (Antony Raj, 2025). This approach helps reveal patterns that are not easily visible in the time domain.

The Laplace transform is a mathematical tool commonly used in signal processing to convert time-domain signals into the complex frequency domain. Compared to basic Fourier analysis, the Laplace transform provides a more general representation that can handle transient and dynamic signals. In ECG signal analysis, the Laplace transform can be used to support both time- and frequency-domain analysis, making it

suitable for studying biological signals (Antony Raj, 2025).

This study focuses on the analysis of ECG signals in both the time and frequency domains using the Laplace transform. The goal is to present a simple and educational approach to ECG signal processing that can help students understand basic biomedical signal analysis concepts (Hal, 2010; MDPI, 2022; Antony Raj, 2025).

II. METHODOLOGY

This section explains the practical steps and signal processing techniques used in this project, which focuses on building a simple ECG device using an Arduino and analyzing its signal in both time and frequency domains.

2.1 ECG Signal Acquisition with Arduino

The ECG signal is captured using a low-cost ECG sensor module connected to an Arduino microcontroller, following the method demonstrated in the “Simplest ECG using Arduino” YouTube tutorial.

Electrodes are placed on the user’s body, and the sensor measures electrical potential differences produced by the heart. These analog signals are then sent to the Arduino’s analog input pin for reading. This setup allows basic ECG waveform acquisition without medical-grade equipment, making it suitable for educational purposes.

Depending on the hardware used (e.g., an AD8232 ECG sensor module), the ECG signal is typically conditioned by the sensor before reaching the Arduino. The sensor amplifies and filters the small electrical signals from the body so that the Arduino’s built-in ADC (Analog-to-Digital Converter) can read the signal reliably. The sensor output is sampled repeatedly in time via the Arduino code, which reads voltage values and sends them to the computer for visualization.

2.2 Pre-Processing of ECG Signals

Raw ECG signals from an Arduino are often noisy due to electrical interference, muscle

movement, and baseline drift. To improve the signal quality, basic preprocessing is performed:

- **Filtering:** Although many beginner Arduino projects (e.g., AD8232 based) include hardware filtering in the ECG module itself, software filtering can also be added. Simple digital filters such as low-pass and high-pass filters which help reduce baseline wandering and high-frequency noise that could obscure the ECG waveform.

- **Normalization:** The captured signal values are normalized to a standard range to make comparisons easier across different recordings. These preprocessing steps enhance the clarity of the ECG waveform before further analysis.

2.3 Time-Domain Signal Analysis

Time-domain analysis involves observing the ECG waveform directly. The ECG signal is plotted as amplitude versus time, allowing identification of typical features like peaks (which correspond to heartbeats) and intervals between them. In this project, time-domain analysis helps students visually understand the heart’s electrical activity and observe changes over time, such as heart rate variations.

Basic measurements such as peak detection and interval timing can be implemented once a relatively clean ECG signal is obtained. The Arduino’s serial plotter or external software like Processing or MATLAB may be used to display the waveform and calculate simple metrics.

2.4 Laplace Transform–Based Frequency Interpretation

Beyond the basic time-domain view, this project introduces a foundational signal processing concept: the Laplace transform. The Laplace transform is a mathematical tool that converts a time-domain signal into a complex frequency domain representation, capturing both sinusoidal and transient components of the signal simultaneously. Biboc (2020) and Zemzemi (2009) discussed how transform methods, including the Laplace transform, can provide

deeper insights into non-stationary biomedical signals like ECG.

In this project, a simple Laplace transform is applied to the pre-processed ECG data (after sampling with Arduino) to examine frequency characteristics. This is done using software tools such as MATLAB or Python rather than directly on the Arduino, because microcontroller resources are limited. The transformed data show how different frequency components contribute to the overall ECG waveform, which can help interpret features that are not obvious in the time domain only.

2.5 Visualization and Interpretation

After applying time-domain and Laplace transform analysis, the results are visualized using plotting tools. Time-domain plots show ECG amplitude changes over time, while frequency-domain representations (derived via Laplace) provide insight into dominant signal components. Students can compare these to understand how the heart's electrical activity appears differently in each domain.

2.6 Use of AI for Support

As part of the learning process, AI support (including this explanation) was used to help organize concepts, clarify technical terms, and present the methodology in a structured way. All scientific and technical ideas including signal processing techniques are grounded in established references and practical Arduino implementations.

III. MATERIALS AND MODULES

A. Arduino Uno

The primary microcontroller for this project is the Arduino UNO, which handles all incoming data from the ECG sensor AD8232. The sensor's analog signals are read, transformed into digital data, and then sent to the display unit. This configuration ensures that the electrical activity of the heart is continuously processed in real time.

C. TFT SPI Display

The ECG data is visualized in real time on the TFT SPI display, which provides a graphical depiction of the heart's electrical activity signals. It shows the PQRST waveform, which is essential for understanding the various phases of the cardiac cycle.

The SPI interface connects the display to the Arduino, enabling quick, efficient data transfer. Because of its excellent resolution, even the ECG signal's slight variations are well recorded, providing a thorough view for analysis.

B. ECG Sensor AD8232

A sophisticated bio-potential sensor, the AD8232, is used to detect electrical impulses from the heart. It works by taking out.

To concentrate on the heart's action, weak bio-signals from the body are amplified to a readable level, and unnecessary noise is removed.

This sensor works exceptionally well at separating the tiny electrical impulses connected to the PQRST waveform of the heart, which qualifies it for ECG monitoring.

D. Electrodes

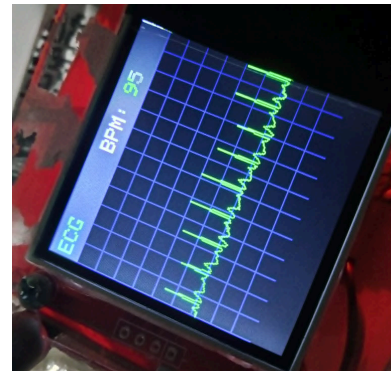
Three electrodes are applied to the left chest, right chest, and right abdomen in this system in order to monitor the electrical signals from the heart. These sites were selected in order to better capture the heart's bio-potential, particularly for the PQRST waveform's vivid display. The AD8232 sensor receives the signals from the electrodes and processes them

E. Resistors

For the system to function smoothly and have clear signals, resistors are necessary. They assist in balancing the impedance of the SPI communication lines that reduce signal deterioration and reflections. Resistors also reduce crosstalk between signals and filter out high-frequency noise, both of which are crucial for capturing sensitive ECG signals.

F. Circuit

The circuit integrates all components—sensor, resistor, Arduino, and screen—working together to capture, process, and display ECG data. The AD8232 sensor collects heart signals, which are then filtered and amplified, while resistors minimize noise. These signals are transmitted to the Arduino UNO for processing, and the resulting PQRST waveform is displayed on the TFT SPI screen in real-time, providing a clear output for continuous monitoring.



IV. HARDWARE AND SOFTWARE APPLICATION

The system uses an Arduino Uno, AD8232 ECG sensor, electrodes, and a TFT SPI display to capture and show ECG signals. The AD8232 sensor reads the heart's electrical activity through the electrodes and sends the signal to the Arduino through pin A0. The Arduino also checks the lead-off pins to detect if the electrodes are disconnected.

For the software, the Arduino code reads the ECG signal, plots the waveform on the TFT display, and shows the estimated BPM using simple peak detection. The signal is also sent to the Serial Monitor, which can be used for further analysis in MATLAB or Python, such as filtering, normalization, and Laplace transform analysis.

V. EXPECTED RESULTS

The expected result of the project is that the system will be able to capture and display the ECG signal in real time using the Arduino, AD8232 sensor, and TFT display. The waveform should show visible heartbeat patterns, especially the repeated peaks that represent the heart's electrical activity.

VI. CONCLUSION

In conclusion, the project successfully shows a simple ECG monitoring system using Arduino and AD8232 sensor. It can display the ECG waveform, estimate BPM, and detect disconnected electrodes. Although it is not for medical diagnosis, it is useful for learning ECG signal acquisition and basic signal analysis.

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